



Bhairav Engine

A STRUCTURED BIO-DESIGN WORK TOOL

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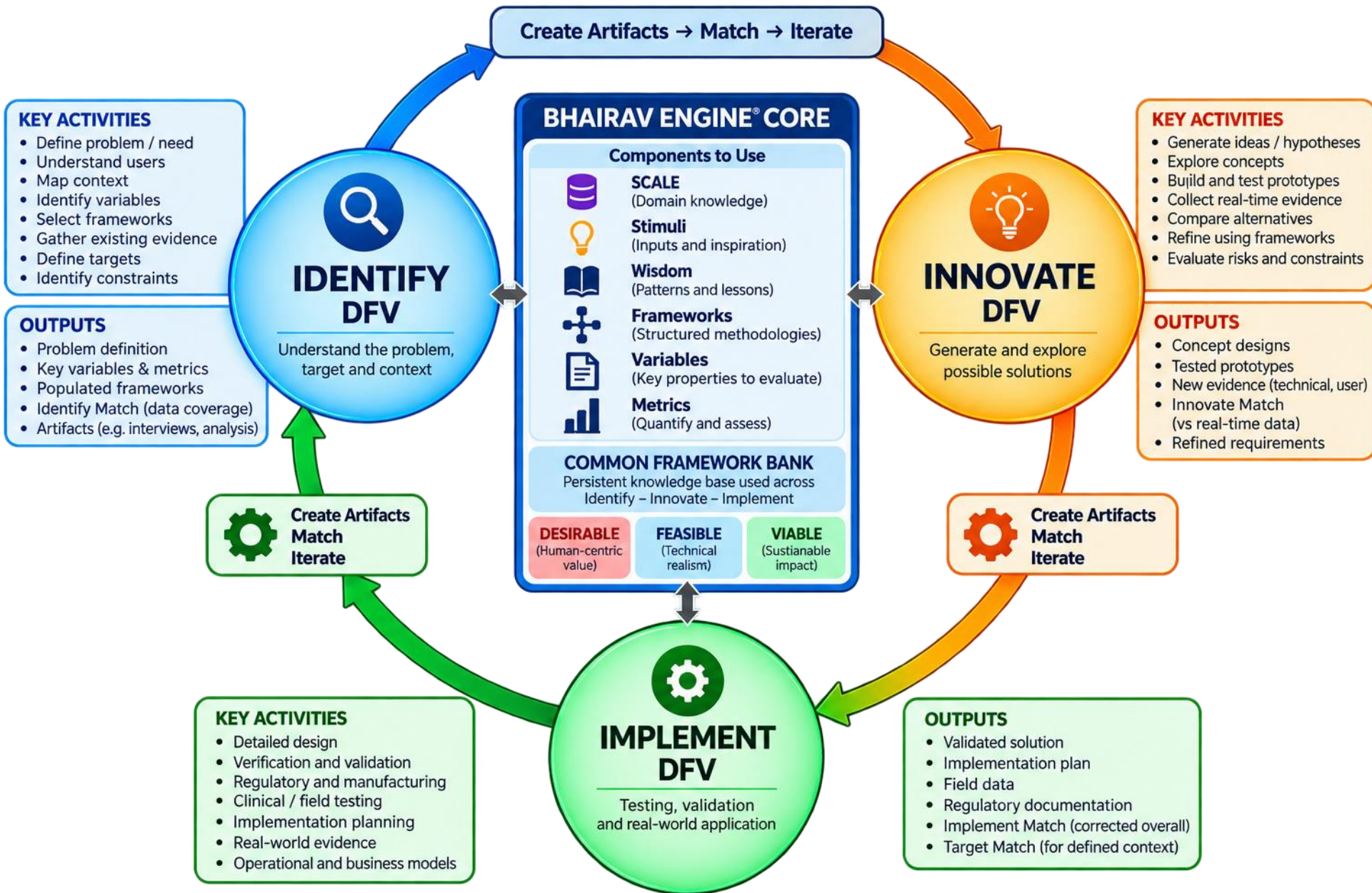
Introduction

- Why use Bhairav Engine? Mainly because cut-throat innovation in medical sciences requires knowledge, skill sharing, and resource support, including human, financial, intellectual, regulatory, marketing, and many other resources, along with ethics and safety.
- In this era of AI and digital health, there is a need for innovators to have a tool through which they can manage all these aspects in a single workspace, with analytics and decision-making support. Making a medical device now requires multiple mindsets and a broader approach compared to the past.
- Bhairav Engine aims to bring these different aspects together in a structured workspace, helping innovators move from identifying a problem to developing, evaluating, and translating an idea into a meaningful and responsible solution.

Bhairav Engine Principle

- Bhairav Engine's principle lies in structuring data, converting it into numerical values, and matching it with your desired targets or industry standards. On this simple principle, it operates.
- To enable this principle, it has several components that help structure and analyse data related to your innovation journey across all phases.

Core Architecture



Core Components



SCALE

Domain-specific scope and level of application for the innovation.



STIMULI

Inputs, triggers and inspirations that drive the innovation process.



WISDOM

Patterns, lessons and existing knowledge to guide decisions.



FRAMEWORKS

Structured methodologies to organize variables and metrics.



VARIABLES

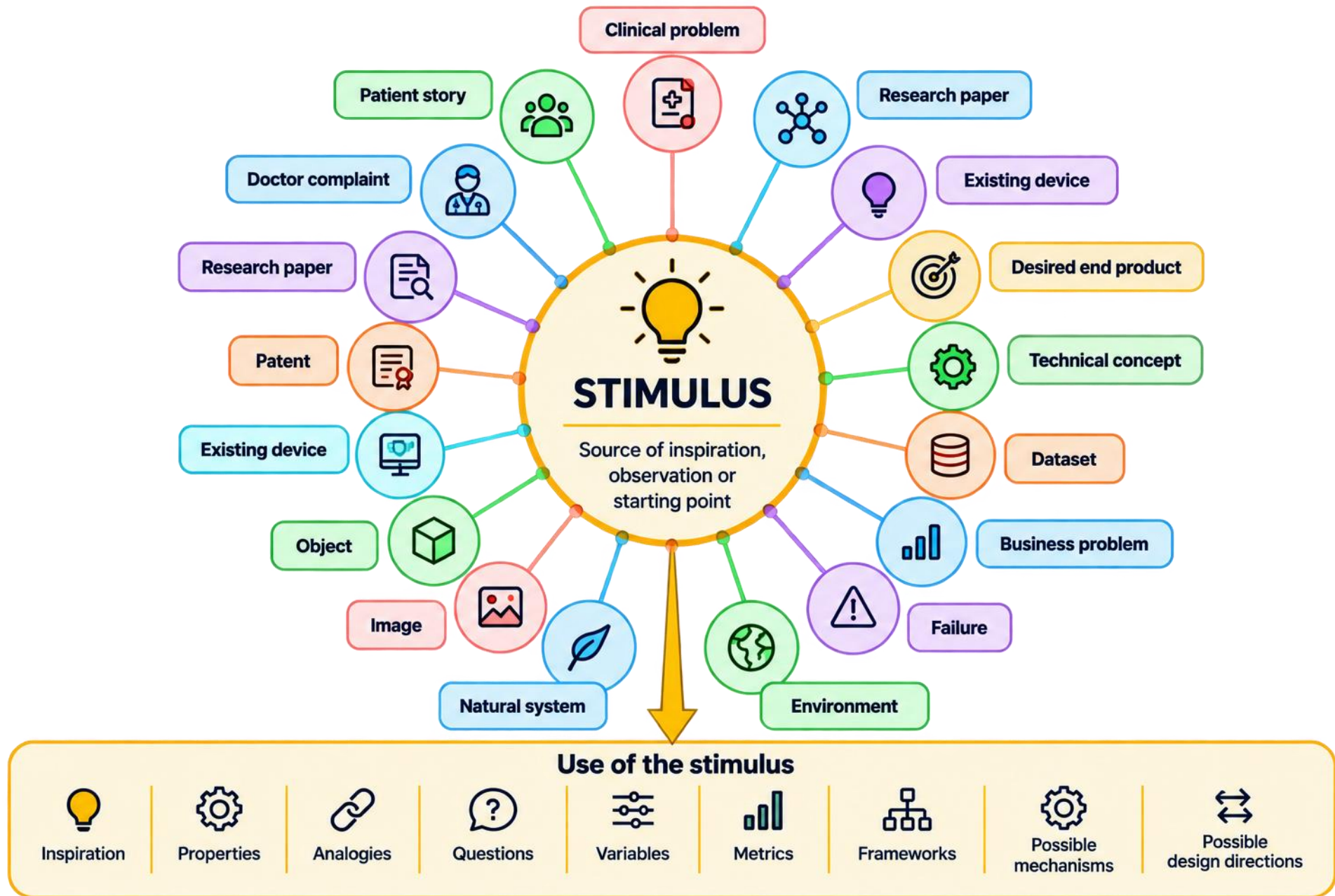
Key properties and factors to be measured or controlled.



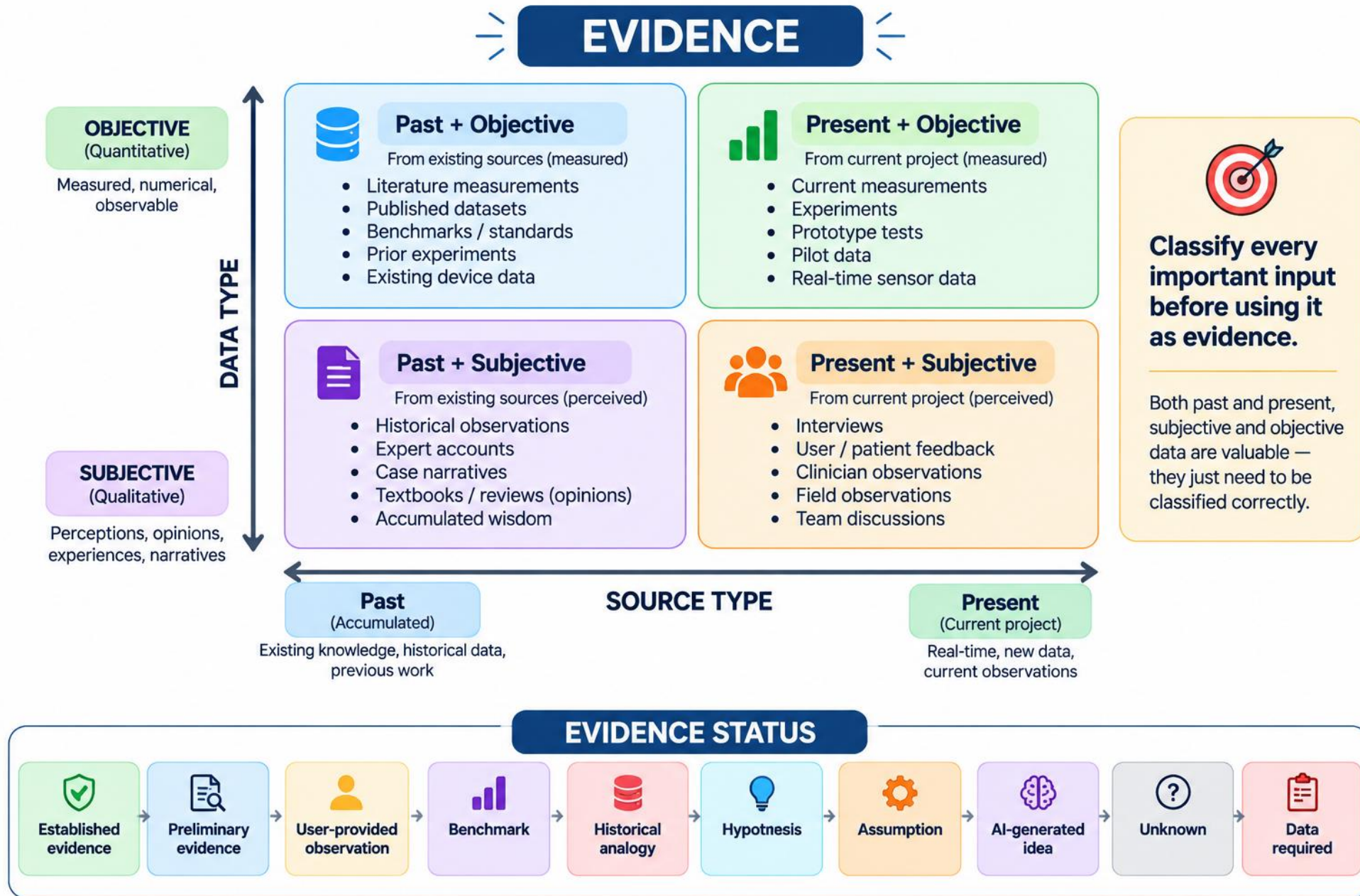
METRICS

Quantitative measures to assess, compare and evaluate outcomes.

Stimulus



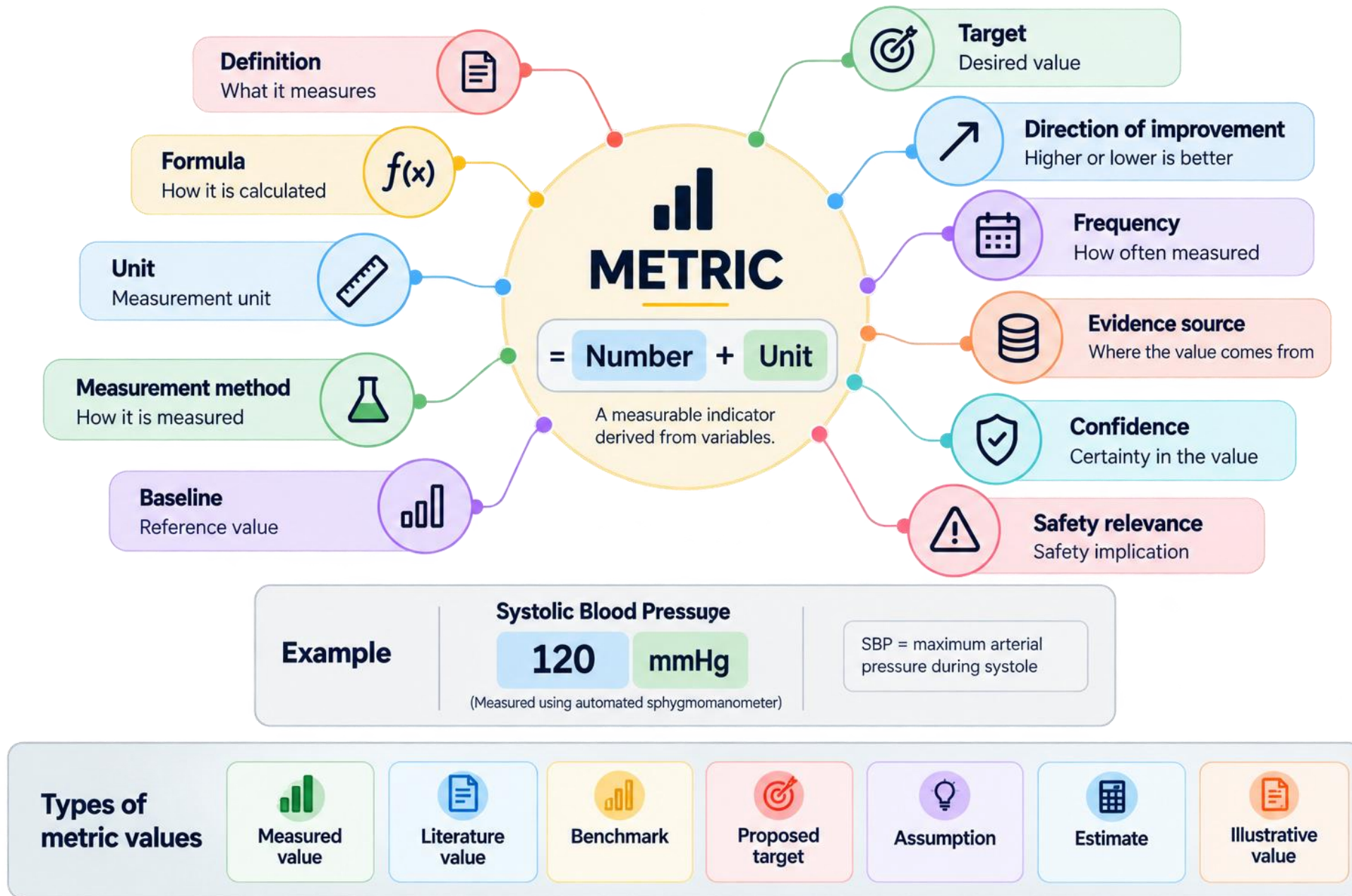
Data or Evidence



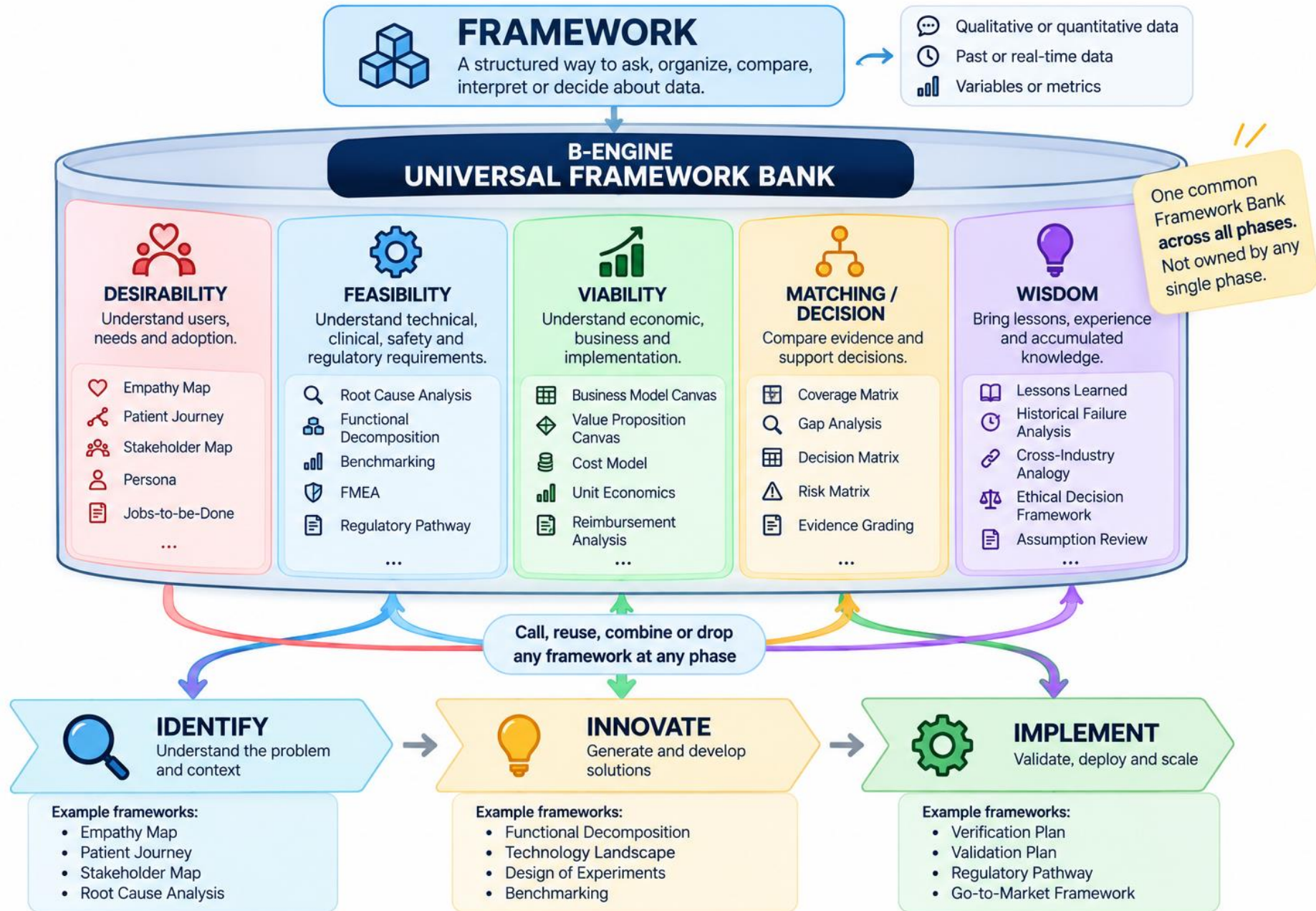
Variable



Metrics



Framework



Scale



SCALE

A bumper pack of frameworks with pre-filled or suggested variables, metrics and supporting knowledge for a particular product, service or problem category.

For a particular product, service or problem category



Medical device



Drug



Diagnostic software



Nutraceutical



Health service

...



SCALE $\neq$ Framework

A Framework is one structured method.

A SCALE is a collection of relevant frameworks and supporting information.



SCALE $\neq$ final answer

A SCALE provides a starting structure.

It must never override project-specific evidence, can only aid to progress faster.

Pick a relevant SCALE



IDENTIFY



INNOVATE



IMPLEMENT



Faster start
Less effort
More structure

Use the same SCALE across all phases
(or call additional ones as needed)

Artifact



ARTIFACT

An artifact is anything the user **creates, fills, modifies, measures or produces** during the **B-ENGINE** process.

Examples of artifacts

- Sketch
- Variable / Metric
- Framework / SCALE
- Interview / Questionnaire
- Dataset
- Diagram
- Prototype
- CAD
- Code
- Simulation
- Experiment
- Report / Publication / Thesis
- Requirement / Architecture
- Business model
- Protocol
- Dashboard
- Decision
- Documentation



Fundamental rule
“If the user creates it as part of the project, it can be treated as an artifact.”

An artifact should exist because it:

- Reduces uncertainty
- Generates evidence
- Tests an assumption
- Clarifies a decision
- Communicates an idea
- Advances validation

Do not generate artifacts merely for appearance.



Attitudes



THE THREE DFV ATTITUDES

Every project can be examined through three persistent attitudes.



DFV is not divided between phases.
All three attitudes can be examined during
IDENTIFY → INNOVATE → IMPLEMENT
continuously.



DESIRABLE

Does someone actually want, need, understand, accept or benefit from this?



User needs
What do users need?



Patient needs
What do patients need?



Clinical needs
What do clinicians need?



Experience
Is it easy to use and acceptable?



Accessibility
Can people access and adopt it?



Stakeholders
Who is involved?



Social value
Does it improve community health?



Public health interest
Is it aligned with government goals?



FEASIBLE

Can it actually be scientifically, technically and operationally achieved?



Science
Is the science sound?



Engineering
Can it be designed and built?



Technology
Are the required technologies available?



Manufacturing
Can it be manufactured at scale?



Performance
Does it meet required performance?



Safety
Is it safe to use?



Regulation
Can it meet regulatory requirements?



Operations
Can it be operated reliably?



VIABLE

Can it sustainably exist, operate and deliver value?



Economics
Is it economically sustainable?



Business model
How will it create and capture value?



Buyer / Payer
Who will pay and buy it?



Procurement
How will it be procured?



Reimbursement
Is reimbursement possible?



Market
Is there a market?



Sustainability
Can it sustain long term?



Policy
Is it aligned with policy?



IDENTIFY

Understand the problem and context



INNOVATE

Generate and develop solutions



IMPLEMENT

Validate, deploy and scale

DFV is continuous across all phases

Examine Desirable, Feasible and Viable at every step.

Phases

THE THREE PHASES

B-ENGINE has three major phases. They are stages of evidence and project maturity. They are not separate framework worlds.



The same DFV attitudes and Framework Bank remain available across all phases. All three attitudes can be examined continuously.



IDENTIFY

Understand the problem, opportunity, target and relevant DFV dimensions.

D

Who has the problem? What is the need?



F

What causes it? What is possible? What are the gaps?



V

Who will buy? Who will pay? What are the constraints?



Key activities

- Understand users and needs
- Identify technical and clinical gaps
- Map stakeholders and ecosystem
- Explore economic and operational context

Identify Match

60%

Coverage of relevant framework fields with real/project data.



INNOVATE

Create and explore possible solutions using the structured information.

D

Will users want and accept this?



F

Can we design and build this?



V

Can it be sustainable?



Key activities

- Generate and explore ideas
- Prototype, test and gather new evidence
- Compare hypotheses with real-time data
- Filter and refine solutions

Innovate Match

75%

Comparison of proposed solution with new real-time evidence.

Refine, iterate and update using new evidence



IMPLEMENT

Validate, deploy and scale the selected solution.

D

Is it effective and accepted in real-world use?



F

Can it be manufactured, safe and compliant?



V

Can it be scaled, reimbursed and sustained?



Key activities

- Regulatory approvals and validation
- Manufacturing and quality systems
- Clinical and field implementation
- Market, procurement and scaling

Implement Match

90%

Re-evaluates using implementation, validation and field data.

1 IDENTIFY
Problem and opportunity

2 INNOVATE
Develop and test solutions

3 IMPLEMENT
Deploy and scale

REAL-WORLD IMPACT

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DFV is continuous across all phases
Examine Desirable, Feasible and Viable at every step using the same frameworks.

Match



MATCH

Measure evidence, not promise.

Match compares your project data with a **defined target, requirement, claim or question** using a valid method. It helps you understand **where you are**, not whether you will succeed.



MATCH IS NOT AUTOMATICALLY “SUCCESS”

A high Match percentage does not mean the project will succeed.

Match must always answer:

Matched against WHAT?

A Match requires:

1



A defined target
What are you trying to meet?

2



A claim, requirement or question
What are you checking?

3



Appropriate evidence
What data or information are used?

4



A valid matching method
Use the simplest valid method.

5



Known uncertainty
Report the limits and confidence.

1

Coverage Match

How much of the framework has been populated?

$$\text{Coverage} = \frac{\text{Completed relevant items}}{\text{Total relevant items}} \times 100$$



Primarily useful during **IDENTIFY**.

2

Threshold Match

Does the actual value meet the predefined requirement?



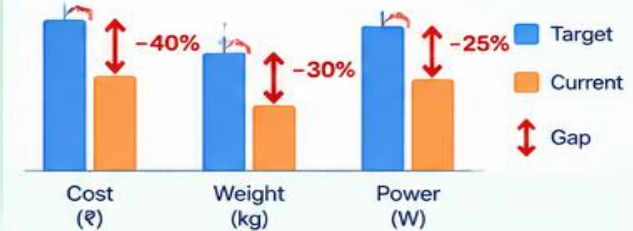
3

Gap Match

Compares current state against target.

$$\text{Gap} = \text{Target} - \text{Actual}$$

(State the direction of improvement)



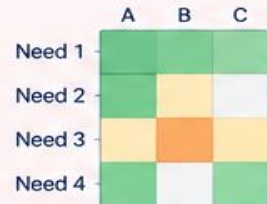
4

Similarity Match

Compare similarity for needs, features, narratives, text etc.

Examples:

- User needs
- Features
- Narratives
- Text
- Requirements



Low similarity High similarity

MATCH

Compare project data with target, requirement or claim using a valid method.

5

Evidence Match

Is the claim supported by appropriate evidence?

- Supported by multiple sources
- Supported by single source
- Limited evidence
- No adequate evidence

6

Requirement Match

Does the solution satisfy defined requirements?

Requirement	Status
Accuracy ≥ 95%	Met
Battery life ≥ 8 h	Met
Weight ≤ 1 kg	Not met
Waterproof (IP67)	Met

7

Field Match

Compare intended performance with actual implementation performance.



Use the simplest valid Match method.

Different questions may require different Match types. Match helps you understand progress and gaps, not guarantee success.

Wisdom



WISDOM

The accumulated learning layer.

Wisdom contains historical lessons, failure cases, expert heuristics, ethical principles, cross-industry analogies, previous project lessons and known risks. It can generate ideas and warnings, but it is **not proof**.



What is Wisdom?

The accumulated learning layer from past knowledge and experience, used to generate ideas, provide guidance and give warnings.



What Wisdom is NOT

- ❌ Not proof.
- ❌ Not a final answer.
- ❌ Not a substitute for real data.
- ❌ Not a guarantee of success.

An analogy must be treated as a hypothesis until appropriately tested.



What can Wisdom do?



Generate ideas
Inspire new ways of thinking.



Provide warnings
Highlight potential risks and pitfalls.



Improve decisions
Bring broader perspective.



Suggest analogies
Connect ideas from other domains.

Examples



Historical lesson: A device failed due to poor battery reliability.



Failure case: User adoption was low due to complex user interface.



Expert heuristic: Start with the simplest version that solves the core problem.



Ethical principle: Ensure safety and equity in access.



Cross-industry analogy: Learn from automotive quality systems (e.g. reliability).



Previous project lesson: Early regulatory engagement reduces delays.



Known risk: Supply chain disruption for critical components.

How Wisdom is used in B-ENGINE



Wisdom Bank
Past knowledge and experience



IDENTIFY

- ✓ Learn from past projects
- ✓ Understand common pitfalls
- ✓ Use relevant analogies

Helps to better understand the problem and context.



INNOVATE

- ✓ Generate new ideas
- ✓ Use expert heuristics
- ✓ Apply cross-industry insights

Inspires and guides solution development.



IMPLEMENT

- ✓ Anticipate risks
- ✓ Apply past lessons
- ✓ Use proven practices

Helps in validation, deployment and scaling.



Output

- ✓ Ideas
- ✓ Warnings
- ✓ Guidance
- ✓ Better informed decisions

Approaches to B-Engine

TWO WAYS TO USE B-ENGINE

You can start with a clear product idea or with a problem/observation.



Both modes use the same components

Stimuli | SCALE | Frameworks | Variables | Metrics
Constraints | Wisdom | Match | DFV | Three Phases



END-PRODUCT-DRIVEN MODE

Use this when you already know what you want to build.

Example:
"I want to build a portable diabetic neuropathy screening device."



1



Define the End Product

What is the product or solution?

2



Identify the intended users and setting

Who will use it? Where will it be used?

3



Search/select an appropriate SCALE

Find a relevant SCALE from the library.

4



Modify the SCALE

Adapt it to your specific product.

5



Identify relevant DFV questions

What needs to be desirable, feasible and viable?

6



Select relevant frameworks

Choose the most useful frameworks.

7



Identify variables and metrics

What will you measure and track?

8



Identify constraints

Technical, regulatory, economic, etc.

9



Start IDENTIFY

Begin populating the selected frameworks.

10



Establish the first Match checkpoint

Check coverage of identified information.

Product idea
+ initial context



B-ENGINE

Structured methodology,
common components,
same workflow.



SCALE



Frameworks



Variables
& Metrics



Wisdom



Match

SAME THREE PHASES FOR BOTH MODES



IDENTIFY

Understand the problem
and context.



INNOVATE

Generate and develop
solutions.



IMPLEMENT

Validate, deploy and scale

Continuous DFV and iterative Match across all phases

Use the same frameworks. Refine with new evidence at each phase.



PROBLEM-DRIVEN MODE

Use this when you have a problem, observation or stimulus.

Example:
"Rural patients have difficulty
accessing nerve conduction testing."



1



Capture the Stimulus

Record the problem, observation or input.

2



Extract properties

What are the key characteristics?

3



Identify variables

What factors are involved?

4



Identify relevant metrics

What will indicate progress or success?

5



Explore Desirable, Feasible and Viable dimensions

Understand the problem from DFV perspectives.

6



Select appropriate frameworks

Choose useful frameworks from the bank.

7



Generate possible opportunity areas

Explore different directions.

8



Generate candidate solutions where justified

Propose and explore solution concepts.

9



Test assumptions

Use real data, literature, experts, etc.

10



Converge toward an End Product

Based on evidence, select and refine a solution.

For referencing example

- For an in-text citation:

“...using the Bhairav Engine methodology.¹”

- Then in the references:

1. Yerra Vishnu Akhil Raj Kumar. Bhairav Engine: A Structured Bio-Design Methodology. India: Copyright Office, Government of India; 2026. Copyright Certificate No. LD-20260186628.

Thank You

Use cases and AI/IDE/Workspace Integrations Powerpoints will be released soon...